
Breast Cancer-Predictive Modeling

Installation

To reproduce this analysis:

- Install the required R packages: tidyverse, caret, e1071, randomForest, pROC and rprojroot

Clone the repository and open the project:

- Clone the repo: git clone https://github.com/your-username/WBC.git
- Open WBC.Rproj in RStudio
- Open WBC_analysis.Rmd and run the code chunks sequentially

Abstract

Breast cancer is one of the most common cancers affecting women worldwide. Early detection significantly improves treatment outcomes, making predictive modeling an important tool in clinical decision-making. This project analyzes the Wisconsin Breast Cancer Diagnostic dataset to explore tumor characteristics and build machine-learning models that classify tumors as benign or malignant. The workflow includes data preprocessing, exploratory data analysis, feature engineering, model training, and performance evaluation. Results demonstrate that several models achieve high accuracy, with strong separation between diagnostic classes based on key predictors such as concavity, perimeter, and area.

Problem Statement

Breast cancer diagnosis often relies on manual interpretation of cell nuclei measurements, which can be time-consuming and subject to human variability. The challenge is to develop a reliable, data-driven model that can accurately classify tumors as benign or malignant using measurable features extracted from digitized images of breast tissue. This project aims to evaluate multiple machine-learning algorithms to determine which approach provides the most accurate and interpretable predictions.

Goals

- Build a reproducible machine-learning workflow for breast cancer classification
- Explore and visualize key predictors that differentiate benign and malignant tumors
- Train multiple classification models and compare their performance
- Identify the most important features contributing to diagnostic accuracy
- Produce a clear, well-documented analysis suitable for academic or professional review

Non-Goals

- Developing a clinical-grade diagnostic tool
- Deploying a real-time prediction system or web application
- Performing deep learning or neural network modeling
- Collecting new medical data or conducting clinical trials
- Providing medical advice or treatment recommendations

Data Description

The dataset contains 569 observations and 32 variables:

- Diagnosis (Benign or Malignant)
- 30 numeric predictors to describe cell nuclei characteristics
- Features grouped into:
  - Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Concave points, Symmetry and Fractal dimension
- Each feature includes:
  - Mean, standard error and worst value

Methods

Data Preparation

- Loaded dataset from `.data` file
- Assigned descriptive column names
- Converted diagnosis to factor
- Checked for missing values
- Scaled numeric features
- Split data into training/testing sets

Exploratory Data Analysis

- Summary statistics
- Class distribution
- Correlation matrix
- Visualization of key predictors

Modeling Approaches

- Logistic Regression
- K-Nearest Neighbors
- Decision Trees
- Random Forest
- Support Vector Machines

Evaluation Metrics

- Accuracy
- Sensitivity & Specificity
- Confusion Matrix
- Cross-validation

Results

- Strong separation between benign and malignant tumors in several predictors
- Models achieved high accuracy (often > 95%)

- Feature importance analysis highlights concavity, perimeter, and area as top predictors
- Random Forest and SVM performed particularly well

References

- UCI Machine Learning Repository: Wisconsin Diagnostic Breast Cancer Dataset
- Hastie, Tibshirani, & Friedman — The Elements of Statistical Learning

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Video

Will include final video presentation here